

BTRFS Install (Kali Unkaputtbar)

 kali.org/docs/installation/btrfs



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Have you ever wished you could travel back in time after deleting that all important customer report or installing a broken driver just before heading into the board meeting?

Well, you better read on, because now you can!

All it takes is to install [Kali Linux 2022.1](https://kali.org/docs/installation/btrfs) or newer with btrfs as file system and to enable snapshotting after installation.

You can even boot into any of your saved snapshots via the boot menu and easily rollback to a previous system state:

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[Overview](#)

[btrfs](#) is a modern Copy on Write (CoW) filesystem for Linux aimed at implementing advanced features such as pooling, snapshots, checksums, and integrated multi-device spanning. In particular, the [snapshot](#) support is what makes btrfs attractive for Kali installations on bare metal. Virtualization solutions such as VMware and VirtualBox provide their own snapshotting functionality and using btrfs in those environments is not really required.

The snapshotting strategy of this walk-through centres around a tool called “snapper” from our friends over at SUSE. Snapper transparently hooks into the apt workflow and automatically creates snapshots before and after any apt operation. This neat little feature allows to easily rollback a system after a botched upgrade.

To top things off, we added [grub-btrfs](#) by [Antynea](#) to automatically add a list of snapshots to the grub boot menu.

[Snapper-gui](#) by [Ricardo Vieira](#) is another great tool we use to make our lives easier.

[Installation Overview](#)

Installing Kali Linux with snapshotting functionality is very similar to a standard installation except that we install it with btrfs as file system.

After the installation, we will install some tools and create a default configuration for snapper.

[Partitioning Scheme](#)

When selecting “btrfs” as file system, the installer will automatically create the following subvolume layout:

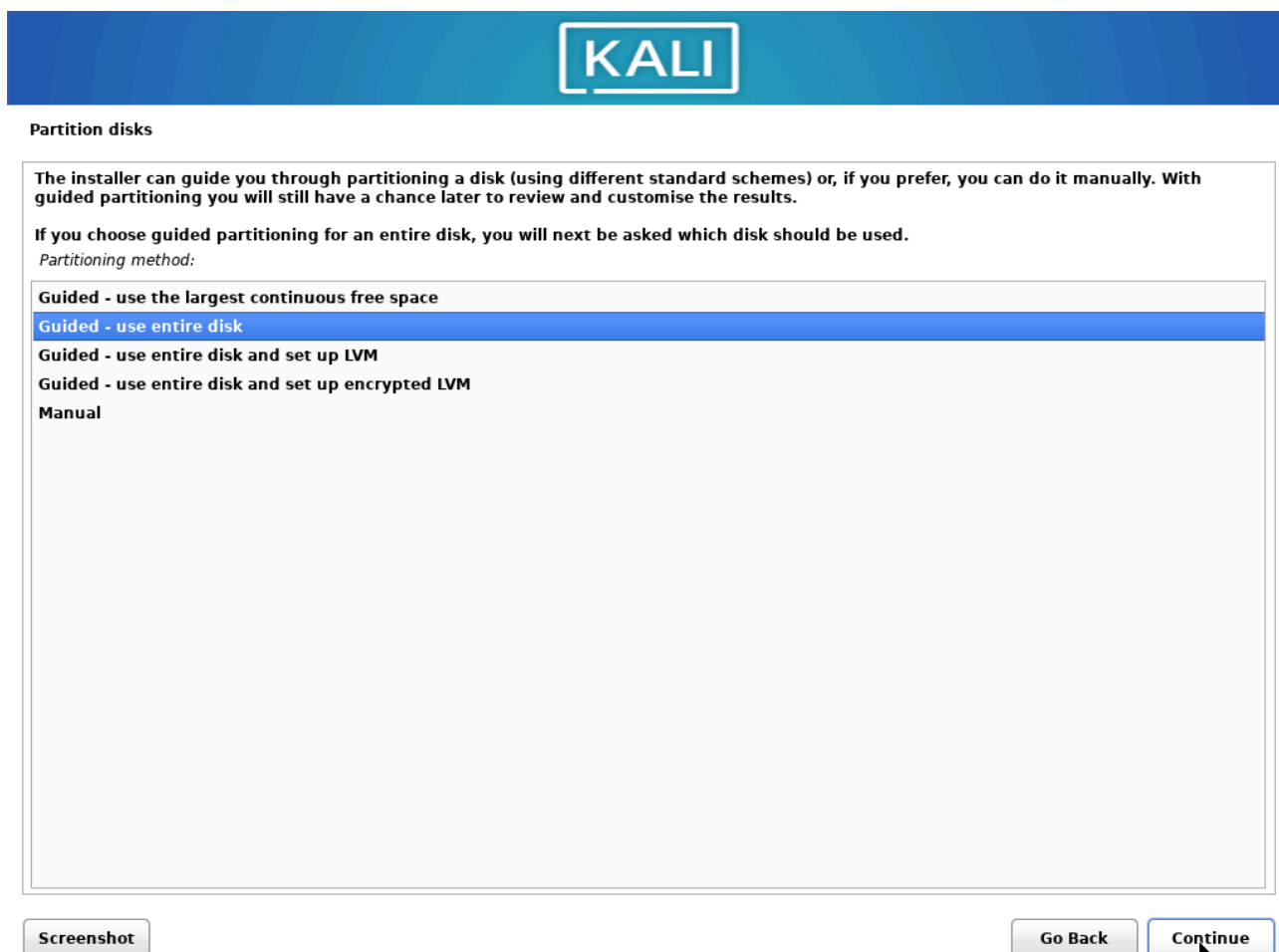
Mount Point	Subvolume	Description
/	@	The root filesystem incl. /boot
/.snapshots	@.snapshots	Snapper's snapshot directory
/home	@home	User home directories
/root	@root	The root user's home directory
/var/log	@var@log	Log files
/srv	@srv	Site-specific data served by this system
/tmp	@tmp	Temporary files
/usr/local	@usr@local	Tertiary hierarchy for local data

Kali Linux BTRFS Installation Steps

Kali Linux Installation Procedure

1. To start your installation, boot with your chosen installation medium. You should be greeted with the Kali Boot screen. Choose *Graphical Install*.
2. The installation steps are identical to a default Kali installation except changing “ext4” to “btrfs” as file system:

At the “Partition Disk” screen, choose “Guided - use entire disk”:



Click “Continue”:

Partition disks

Note that all data on the disk you select will be erased, but not before you have confirmed that you really want to make the changes.

Select disk to partition:

SCSI3 (0,0,0) (sda) - 84.7 GB ATA VBOX HARDDISK

Screenshot

Go Back

Continue

Double-click on the / partition:

Partition disks

This is an overview of your currently configured partitions and mount points. Select a partition to modify its settings (file system, mount point, etc.), a free space to create partitions, or a device to initialize its partition table.

Guided partitioning

Configure software RAID

Configure the Logical Volume Manager

Configure encrypted volumes

Configure iSCSI volumes

SCSI3 (0,0,0) (sda) - 84.7 GB ATA VBOX HARDDISK

>		1.0 MB			FREE SPACE	
>	#1	536.9 MB	B	F	ESP	
>	#2	83.2 GB	F	ext4	/	
>	#3	1.0 GB	F	swap	swap	
>		179.7 kB			FREE SPACE	

Undo changes to partitions

Finish partitioning and write changes to disk

Screenshot

Help

Go Back

Continue

Choose "btrfs":

Partition disks

How to use this partition:

Ext4 journaling file system
Ext3 journaling file system
Ext2 file system
btrfs journaling file system
JFS journaling file system
XFS journaling file system
FAT16 file system
FAT32 file system
swap area
Reserved BIOS boot area
EFI System Partition
physical volume for encryption
physical volume for RAID
physical volume for LVM
do not use the partition

Screenshot

Help

Go Back

Continue

Select "Done setting up the partition" and continue:

Partition disks

You are editing partition #2 of SCSI3 (0,0,0) (sda). This partition is formatted with the btrfs journaling file system. All data in it WILL BE DESTROYED!

Partition settings:

Name:
Use as: btrfs journaling file system

Format the partition: yes, format it
Mount point: /
Mount options: defaults
Label: none
Bootable flag: off

Erase data on this partition

Delete the partition

Done setting up the partition

Screenshot

Help

Go Back

Continue

Select "Finish partitioning and write changes to disk" and continue with the installation.

Partition disks

This is an overview of your currently configured partitions and mount points. Select a partition to modify its settings (file system, mount point, etc.), a free space to create partitions, or a device to initialize its partition table.

Guided partitioning

Configure software RAID

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SCSI3 (0,0,0) (sda) - 84.7 GB ATA VBOX HARDDISK

>	1.0 MB	FREE SPACE
> #1	536.9 MB	B F ESP
> #2	83.2 GB	F btrfs /
> #3	1.0 GB	F swap swap
>	179.7 kB	FREE SPACE

Undo changes to partitions

Finish partitioning and write changes to disk

Screenshot Help

Go Back Continue

3. Login after installation and run the following commands in a terminal:

```
# Set a secure root password or you'll struggle to log into a recovery shell
$ sudo passwd

# Install some essential tools
$ sudo apt update && sudo apt install btrfs-progs snapper snapper-gui grub-btrfs

# Create the snapper configuration for the root filesystem "/"
$ sudo cp /usr/share/snapper/config-templates/default /etc/snapper/configs/root
$ sudo sed -i 's/^SNAPPER_CONFIGS="\\"/SNAPPER_CONFIGS="\\"root\\"/'
/etc/default/snapper

# Prevent "updatedb" from indexing the snapshots, which would slow down the system
$ sudo sed -i '/# PRUNENAMES=/ a PRUNENAMES = ".snapshots"' /etc/updatedb.conf
```

4. We need to tweak the desktop managers to work in read only snapshots. Pick your DE from the following:

GNOME:

```
# Reconfigure gdm to allow booting into read-only snapshots
# GDM needs to have write access to "/var/lib/gdm3" and "/var/lib/AccountService"
during login.
# We have to create additional subvolumes for them:
```

```

$ mount # Pick your main partition, </dev/sda1> in our example, replace </dev/sda1>
it with yours
$ sudo mount </dev/sda1> /mnt
$ sudo btrfs subvolume create /mnt/@var@lib@gdm3
$ sudo btrfs subvolume create /mnt/@var@lib@AccountsService

$ sudo mv /var/lib/gdm3/* /var/lib/gdm3/. * /mnt/@var@lib@gdm3
$ sudo mv /var/lib/AccountsService/* /var/lib/AccountsService/. *
/mnt/@var@lib@AccountsService/

$ sudo vi /etc/fstab # Add the following (substitute the <UUID> with yours)

# /var/lib/gdm3 was on /dev/sda1 during installation
UUID=<dc1ca012-9349-4fcf-b761-ca323379b019> /var/lib/gdm3    btrfs
defaults,subvol=@var@lib@gdm3 0          0

# /var/lib/AccountsService was on /dev/sda1 during installation
UUID=<dc1ca012-9349-4fcf-b761-ca323379b019> /var/lib/AccountsService    btrfs
defaults,subvol=@var@lib@AccountsService 0          0

# Reboot for the changes to take effect
$ sudo reboot

```

KDE:

```

# KDE works out of the box, just reboot and enjoy
$ sudo reboot

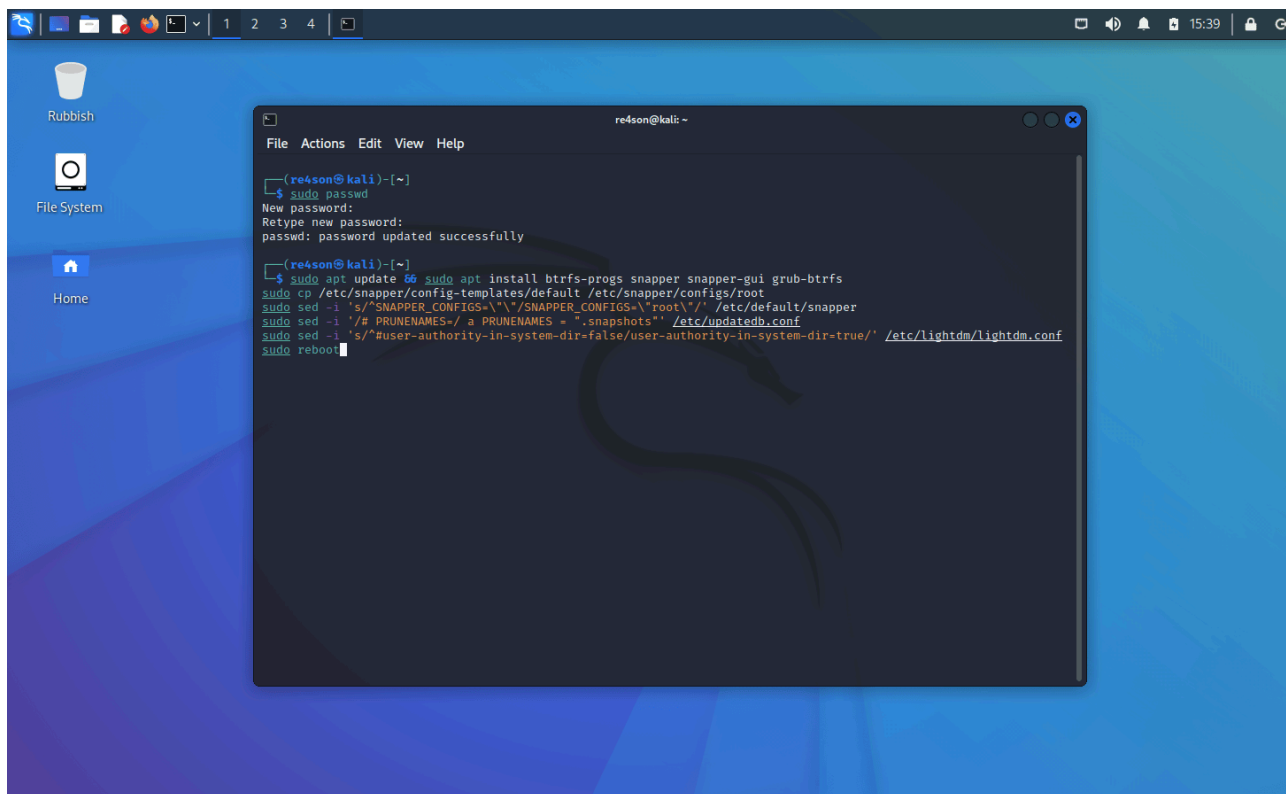
```

XFCE:

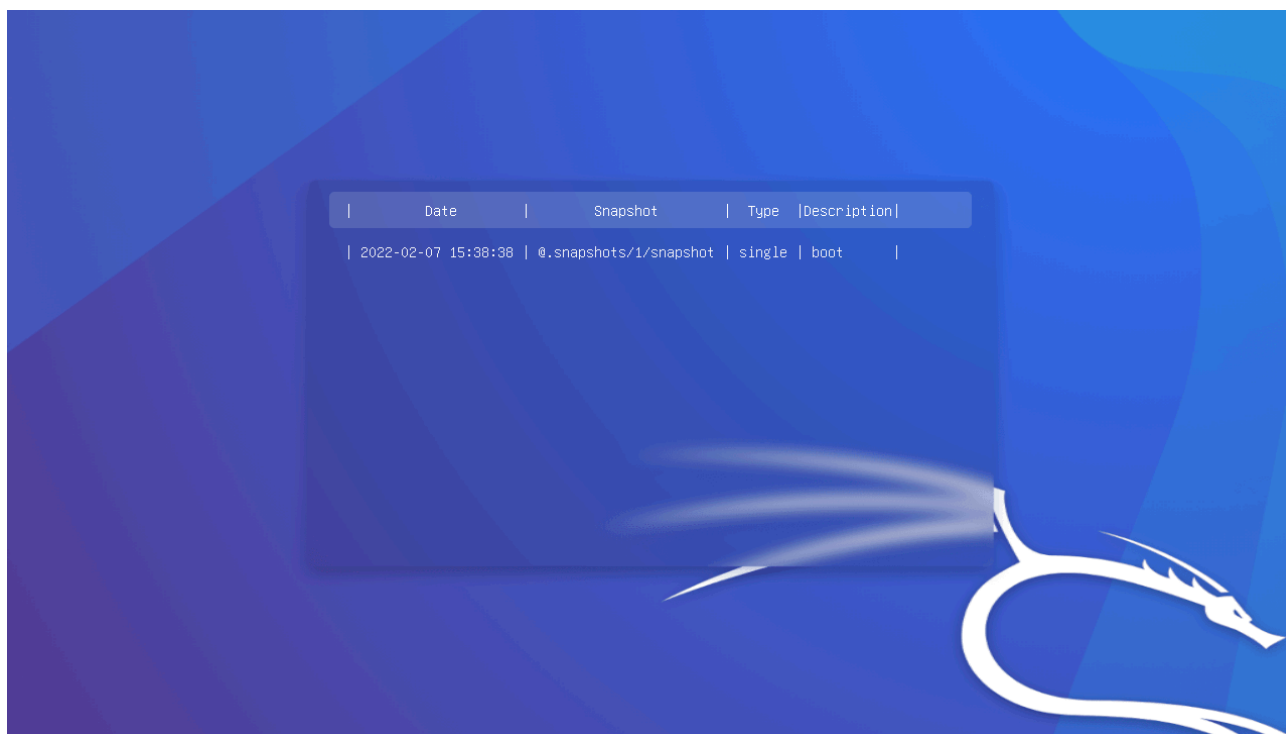
```

# Reconfigure lightdm to allow booting into read-only snapshots
$ sudo sed -i 's/^#user-authority-in-system-dir=false/user-authority-in-system-
dir=true/' /etc/lightdm/lightdm.conf
$
$ sudo reboot

```



5. The first reboot will create the first automatic snapshot. Reboot again to find the new boot menu entry for this snapshot:

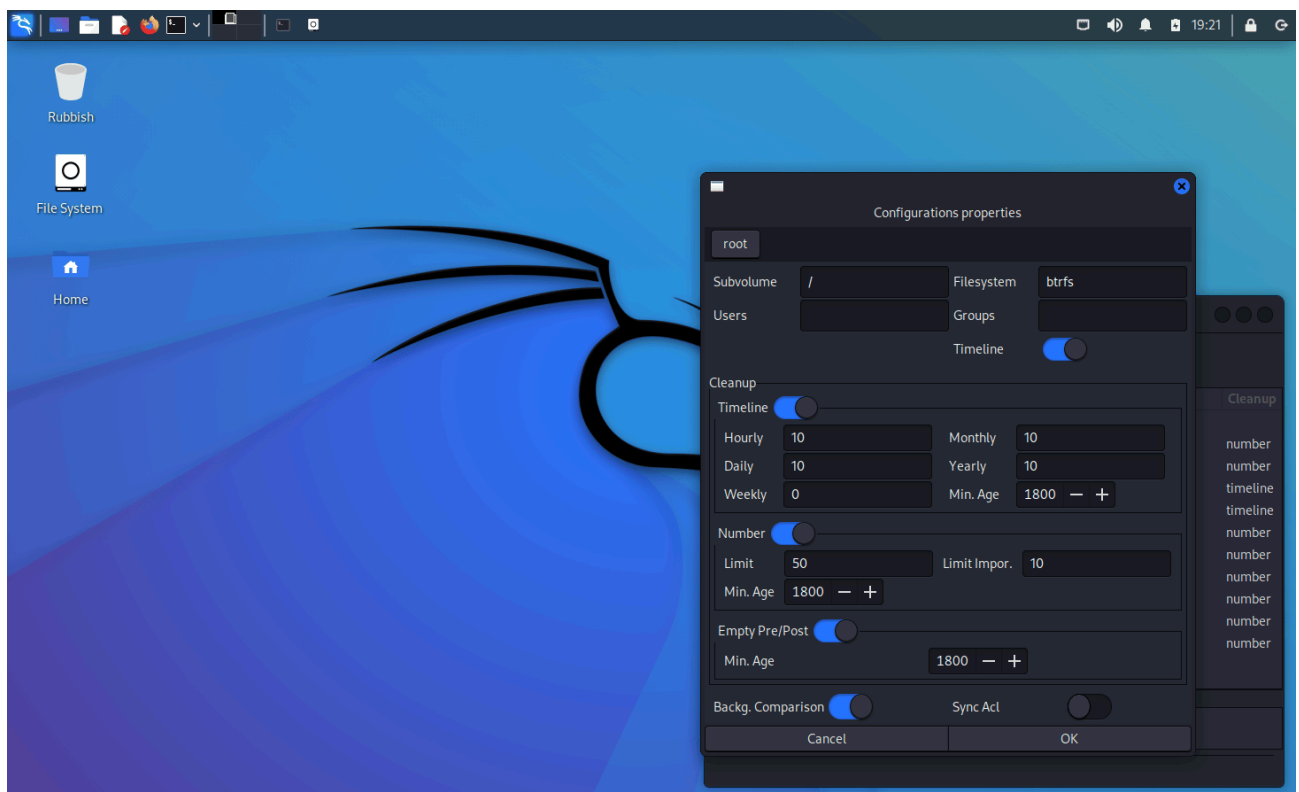
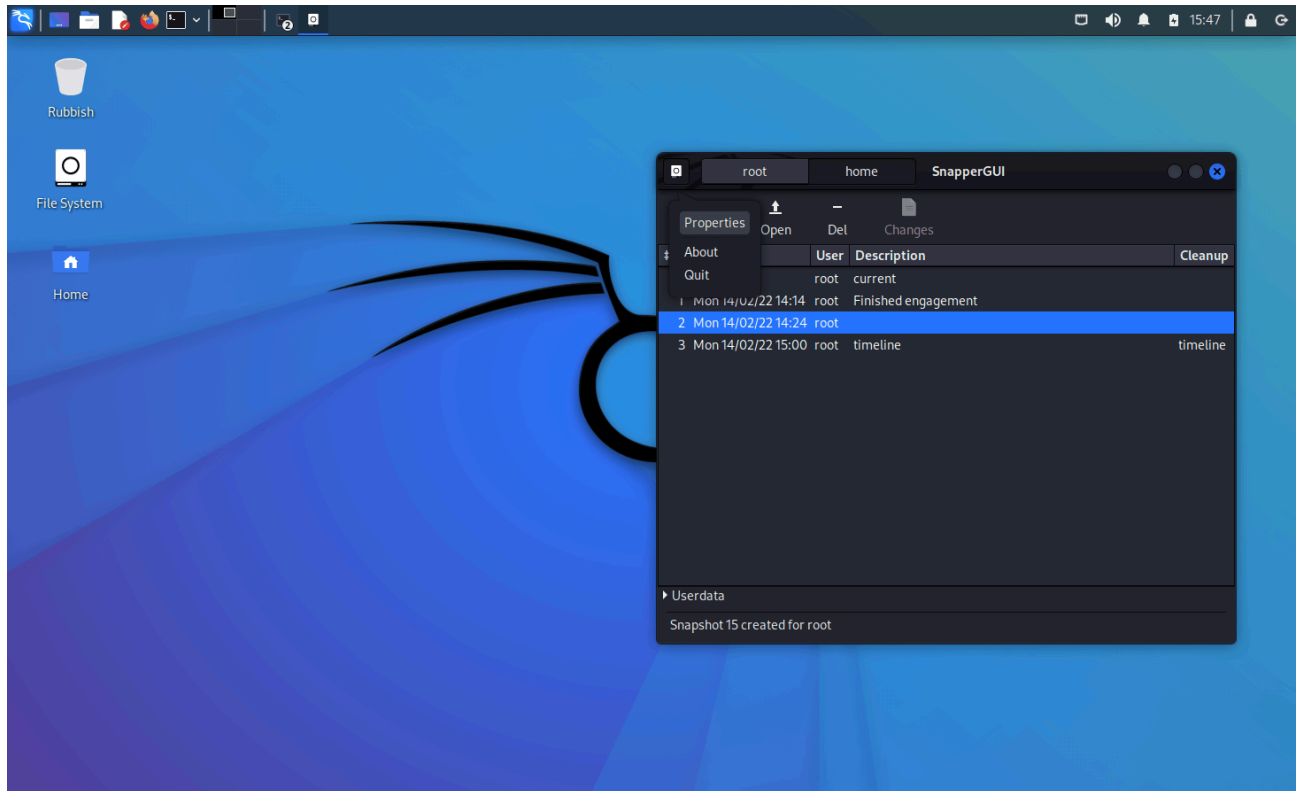


Congratulations, you have just installed a Kali system with automatic snapshotting functionality! Next, we will cover some basic usage examples.

Usage

Modify configurations

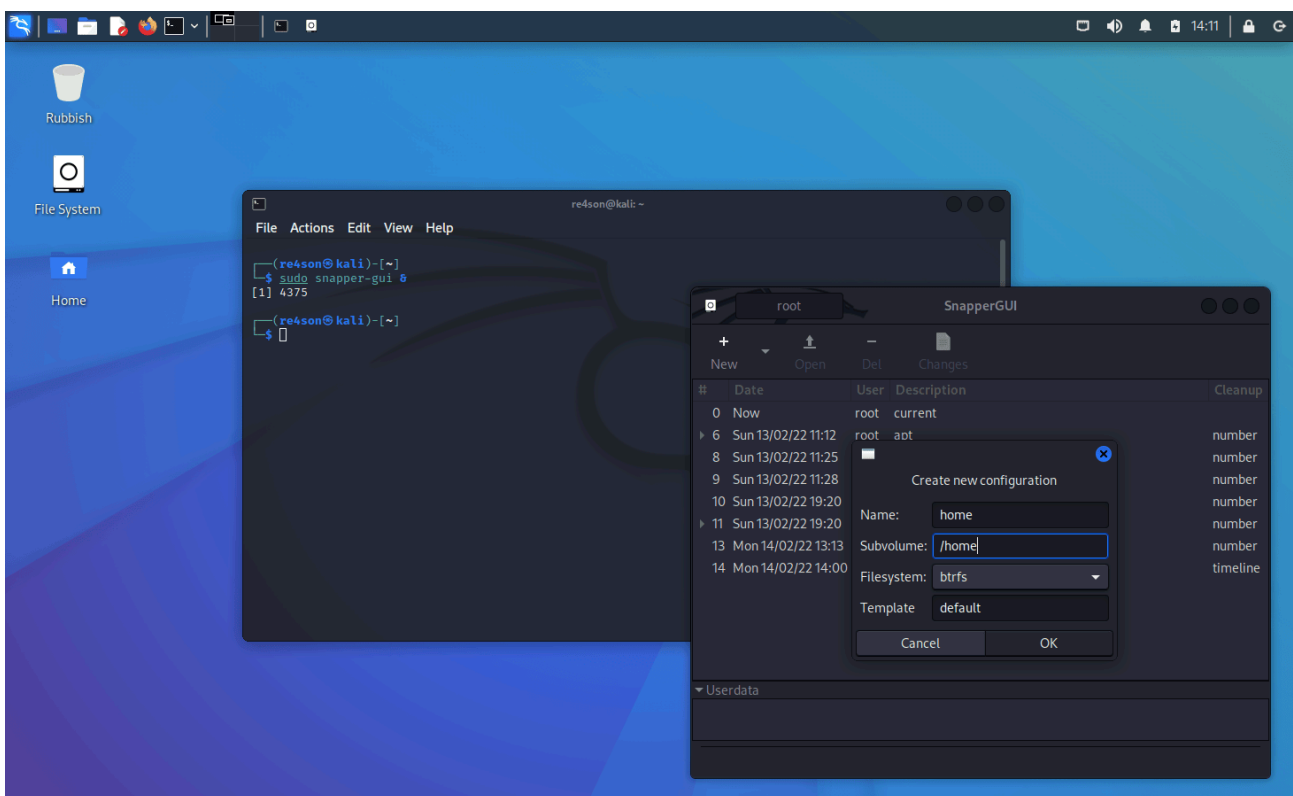
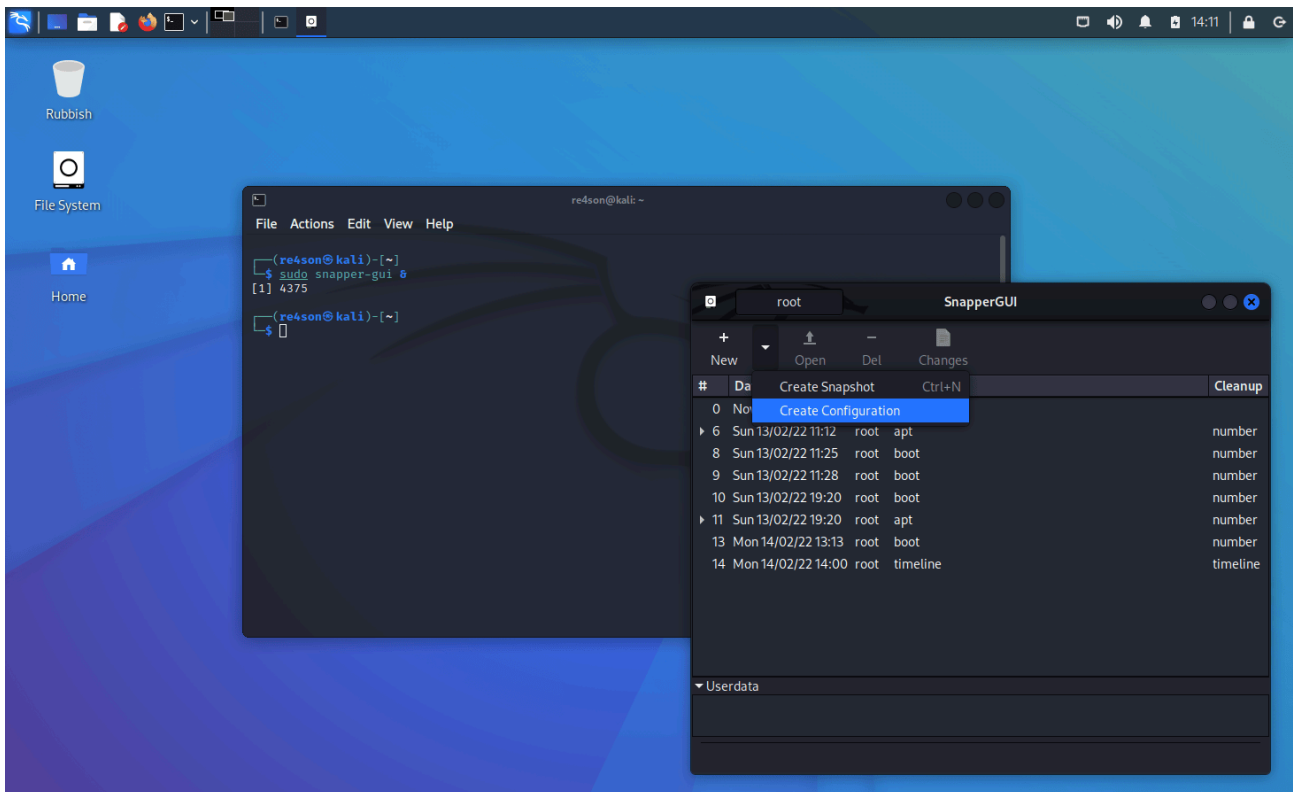
Out of the box Kali creates snapshots of the root directory to allow system rollbacks. Snapshots are automatically created during apt operations, at specified time intervals, and on every boot. The configuration can be changed via the “snapper-gui” tool. Just click on the little icon in the top left hand corner and select “Properties”:



Create additional configurations

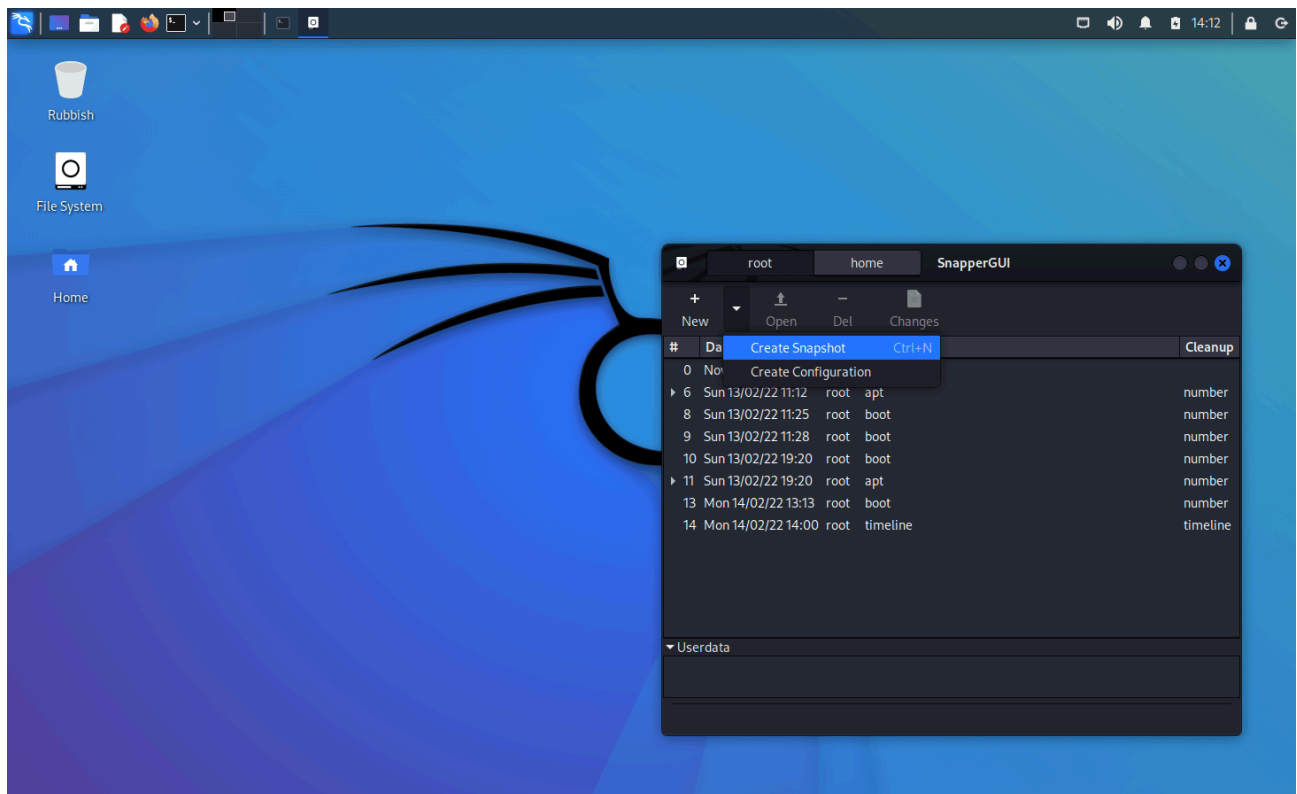
To create snapshots of your home directory, you can create a new configuration using snapper-gui.

Click on “New” -> “Create Configuration”



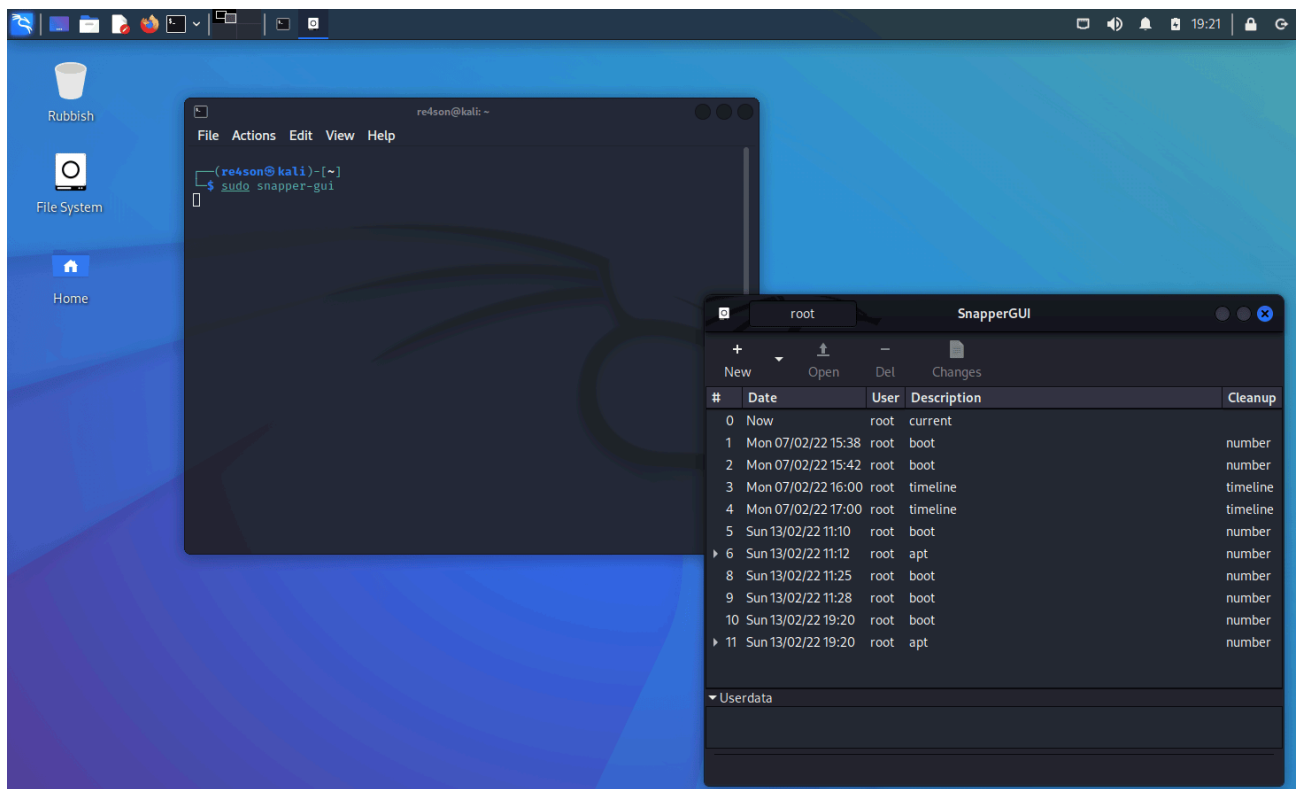
Create a snapshot

To manually create a snapshot using snapper-gui, select the appropriate configuration tab (home in this case) and click “New” -> “Create Snapshot”



List snapshots

Snapshots are listed in the snapper-gui:



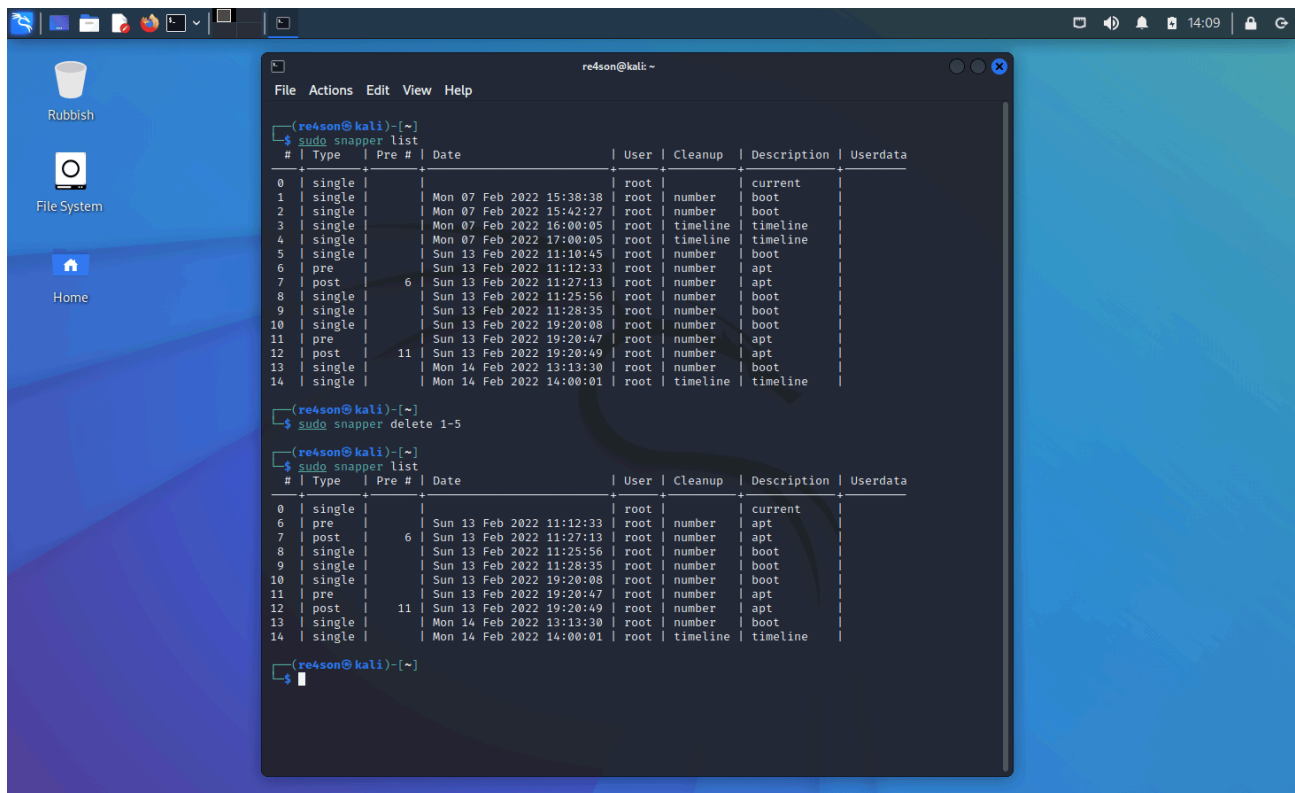
Alternatively, snapshots on all configurations can be viewed using the snapper command line tool:

```
$ sudo snapper list -a
```

Delete snapshots

The easiest way to delete a snapshot is by using the snapper command line tool:

```
sudo snapper delete <number-or-number-range>
```



The screenshot shows a terminal window on a Kali Linux desktop. The user runs `sudo snapper list`, which displays a table of snapshots. Then, the user runs `sudo snapper delete 1-5`, and runs `sudo snapper list` again to show the updated list of snapshots.

#	Type	Pre #	Date	User	Cleanup	Description	Userdata
0	single			root		current	
1	single		Mon 07 Feb 2022 15:38:38	root	number	boot	
2	single		Mon 07 Feb 2022 15:42:27	root	number	boot	
3	single		Mon 07 Feb 2022 16:00:05	root	timeline	timeline	
4	single		Mon 07 Feb 2022 17:00:05	root	timeline	timeline	
5	single		Sun 13 Feb 2022 11:10:45	root	number	boot	
6	pre		Sun 13 Feb 2022 11:12:33	root	number	apt	
7	post	6	Sun 13 Feb 2022 11:27:13	root	number	apt	
8	single		Sun 13 Feb 2022 11:25:56	root	number	boot	
9	single		Sun 13 Feb 2022 11:28:35	root	number	boot	
10	single		Sun 13 Feb 2022 19:20:08	root	number	boot	
11	pre		Sun 13 Feb 2022 19:20:47	root	number	apt	
12	post	11	Sun 13 Feb 2022 19:20:49	root	number	apt	
13	single		Mon 14 Feb 2022 13:13:30	root	number	boot	
14	single		Mon 14 Feb 2022 14:00:01	root	timeline	timeline	

#	Type	Pre #	Date	User	Cleanup	Description	Userdata
0	single			root		current	
6	pre		Sun 13 Feb 2022 11:12:33	root	number	apt	
7	post	6	Sun 13 Feb 2022 11:27:13	root	number	apt	
8	single		Sun 13 Feb 2022 11:25:56	root	number	boot	
9	single		Sun 13 Feb 2022 11:28:35	root	number	boot	
10	single		Sun 13 Feb 2022 19:20:08	root	number	boot	
11	pre		Sun 13 Feb 2022 19:20:47	root	number	apt	
12	post	11	Sun 13 Feb 2022 19:20:49	root	number	apt	
13	single		Mon 14 Feb 2022 13:13:30	root	number	boot	
14	single		Mon 14 Feb 2022 14:00:01	root	timeline	timeline	

Voilà, easy.

Now is probably the right time to flick through the snapper command line options via:

```
snapper --help
```

Rollback

To roll back to a previous snapshot we have to remember two things:

- The root / of the file system has been installed in a subvolume /@" and not the root of the btrfs partition /
- A snapshot is treated like just another subvolume except that they are **read-only**
- You can easily boot into your read-only snapshots to find the one you want to roll-back to.

Thus all we have to do is mount the btrfs partition and replace the current root subvolume @ with the last snapshot. To be safe we'll backup the current root (@) subvolume.

E.g.:

```
# get the device that contains your "/" subvolume and remember it for the next step:
```

```
mount | grep 'subvol=/@)'
```

```
# mount your root partition (replace "/dev/sda2" with yours from above):
```

```
sudo mount /dev/sda2 -o subvol=/ /mnt
```

```
# Move the old root away:
```

```
sudo mv /mnt/@ /mnt/@_badroot
```

```
# Roll back to a previous snapshot by creating a read-write copy of it as "@":
```

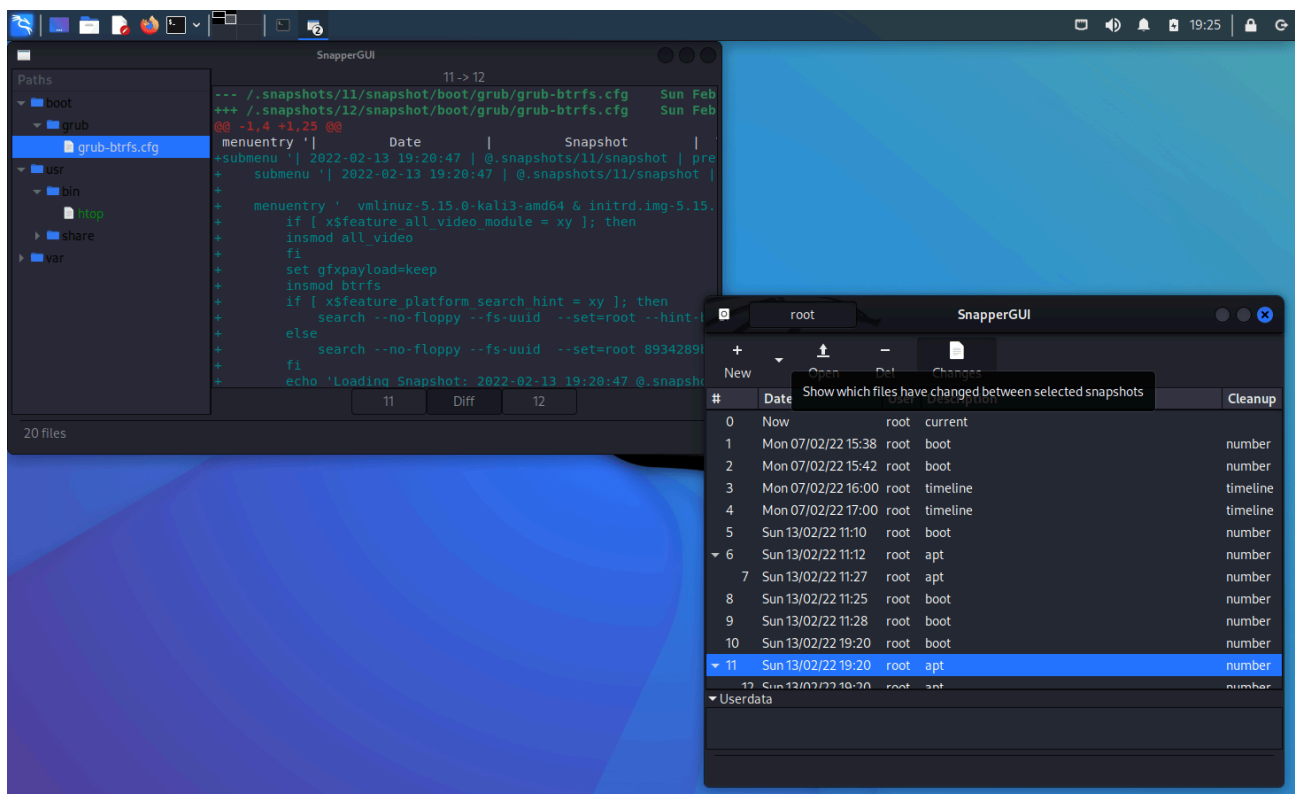
```
sudo btrfs subvolume snapshot /mnt/@.snapshots/XXXXX/snapshot /mnt/@
```

```
# That's it, reboot:
```

```
sudo reboot -f
```

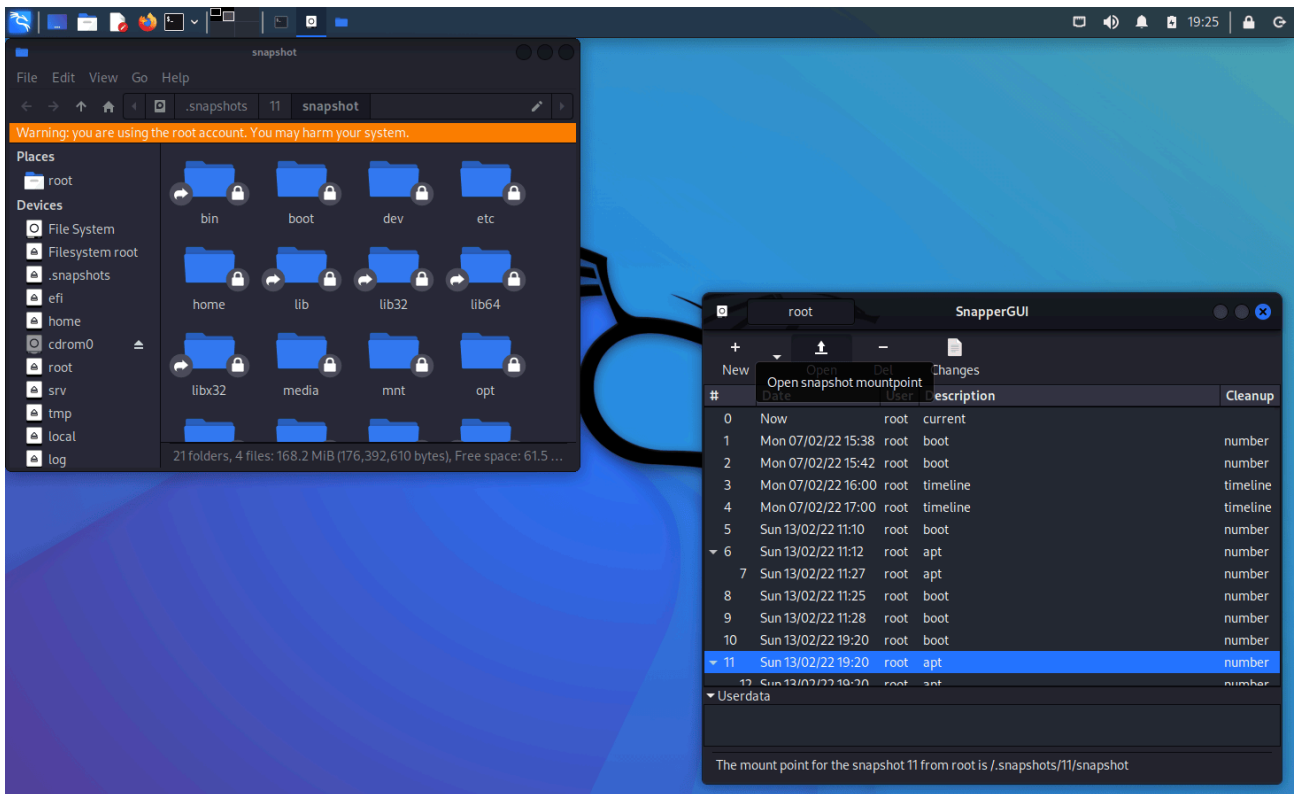
[Diff](#)

Snapper is chock-a-block with powerful features like diffs between snapshots:



[Browse snapshots](#)

You can even browse the content of snapshots:



Full recovery from an unbootable system

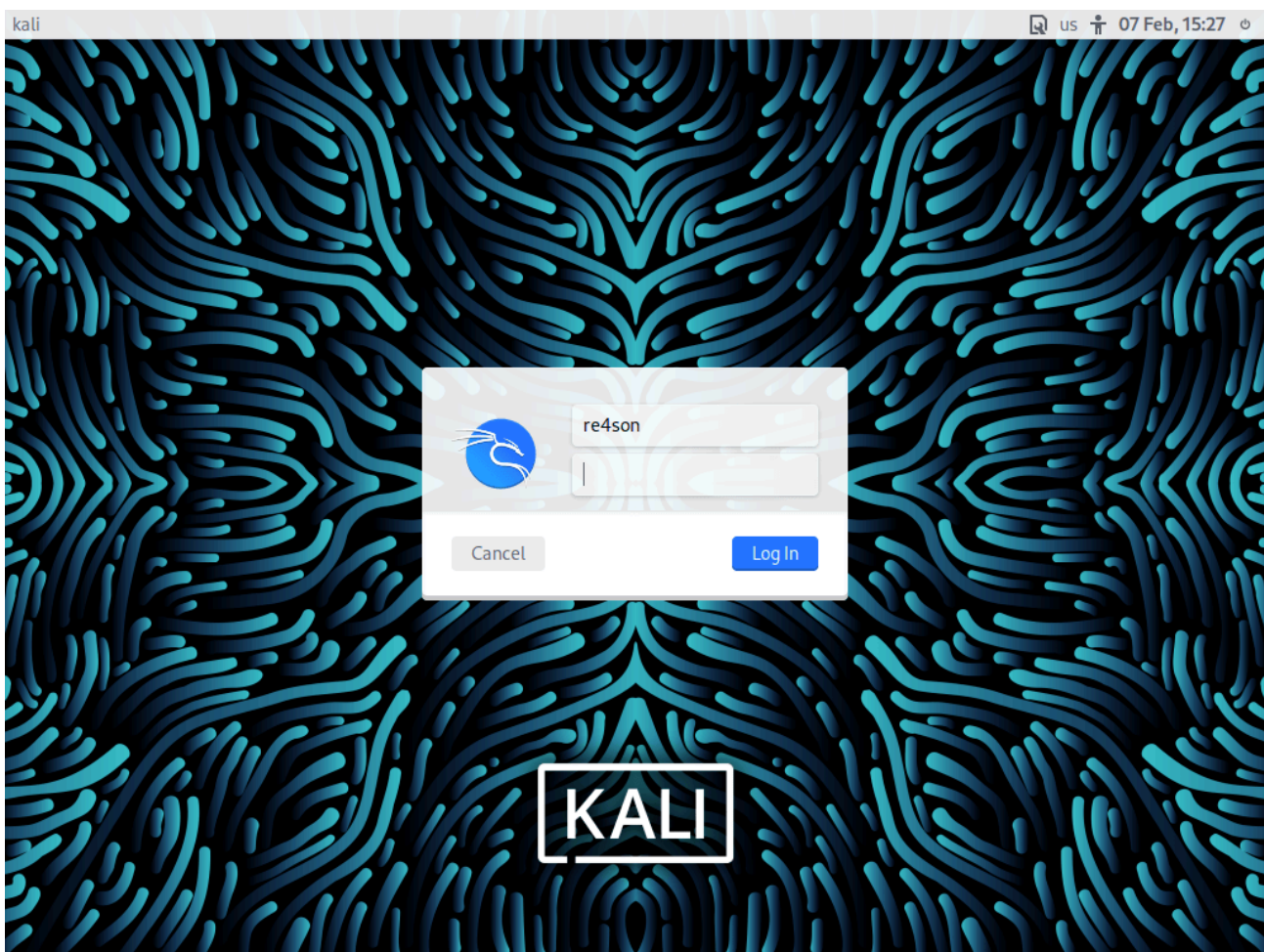
Boot into a last known good snapshot

Let's assume that the last upgrade broke our machine. Every run of "apt install" creates to snapshots, one "pre" snapshot is created before the installation and one "post" snapshot is created after the installation.

To undo the last "apt upgrade", we would boot into the last "pre" snapshot and check if everything is working again:



Now you can login to the snapshot.



Please note that the snapshot is read only and you might receive an error message from an applet or two after logging in. Just ignore that.

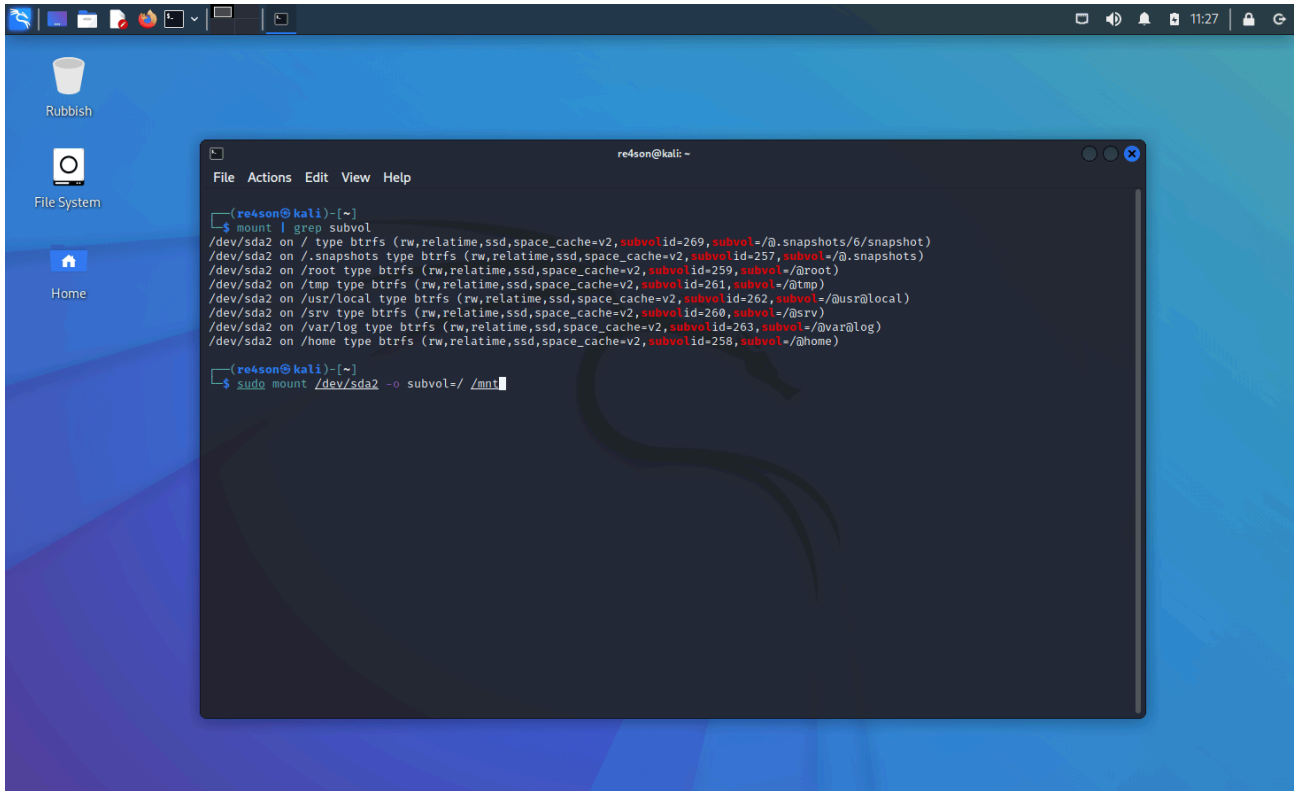
Have a look around ensure that this is what you would like to roll-back to.

Rollback

Remember that “/” itself is the subvolume “@”. To rollback to a snapshot, all we have to do is replace “@” with the snapshot we want.

1. First we have to mount the physical partition that holds all our subvolumes. Let’s find it first

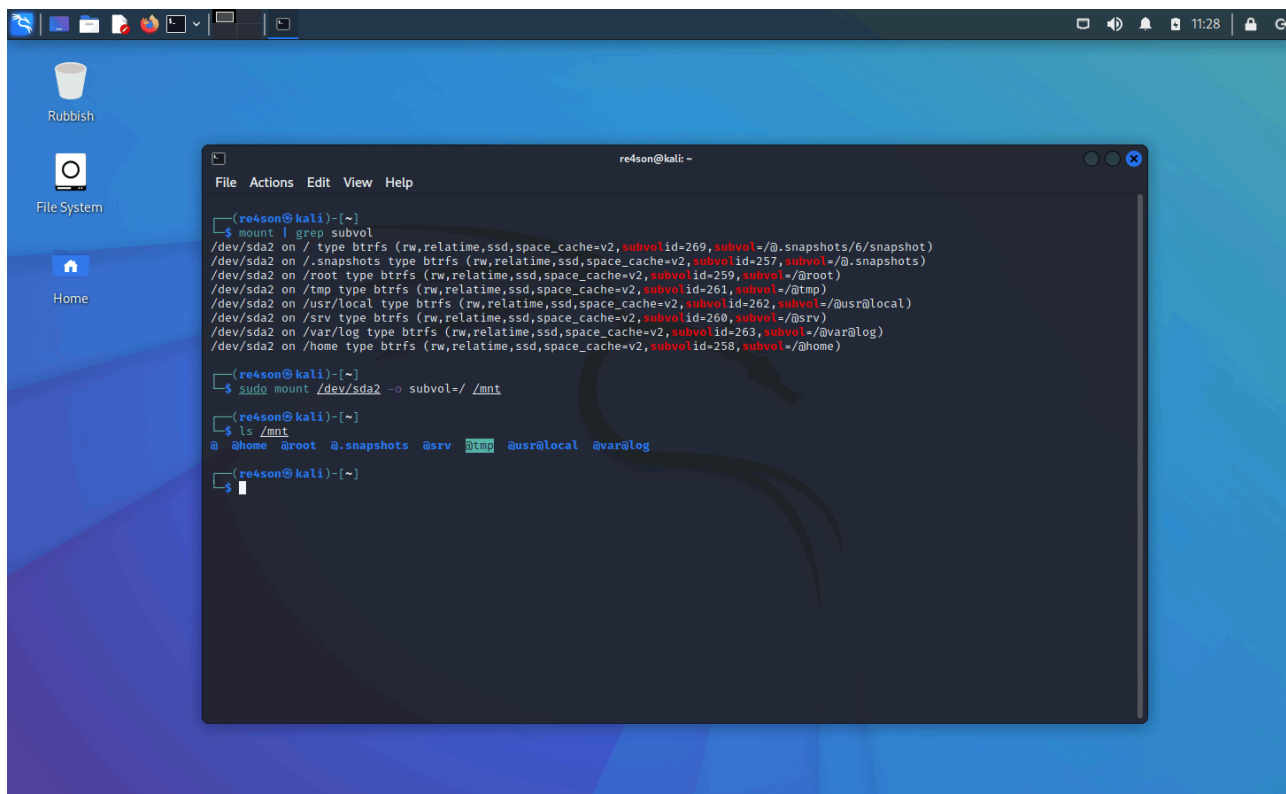
```
mount | grep 'subvol='
```



and then mount it (/dev/sda2 in this example:

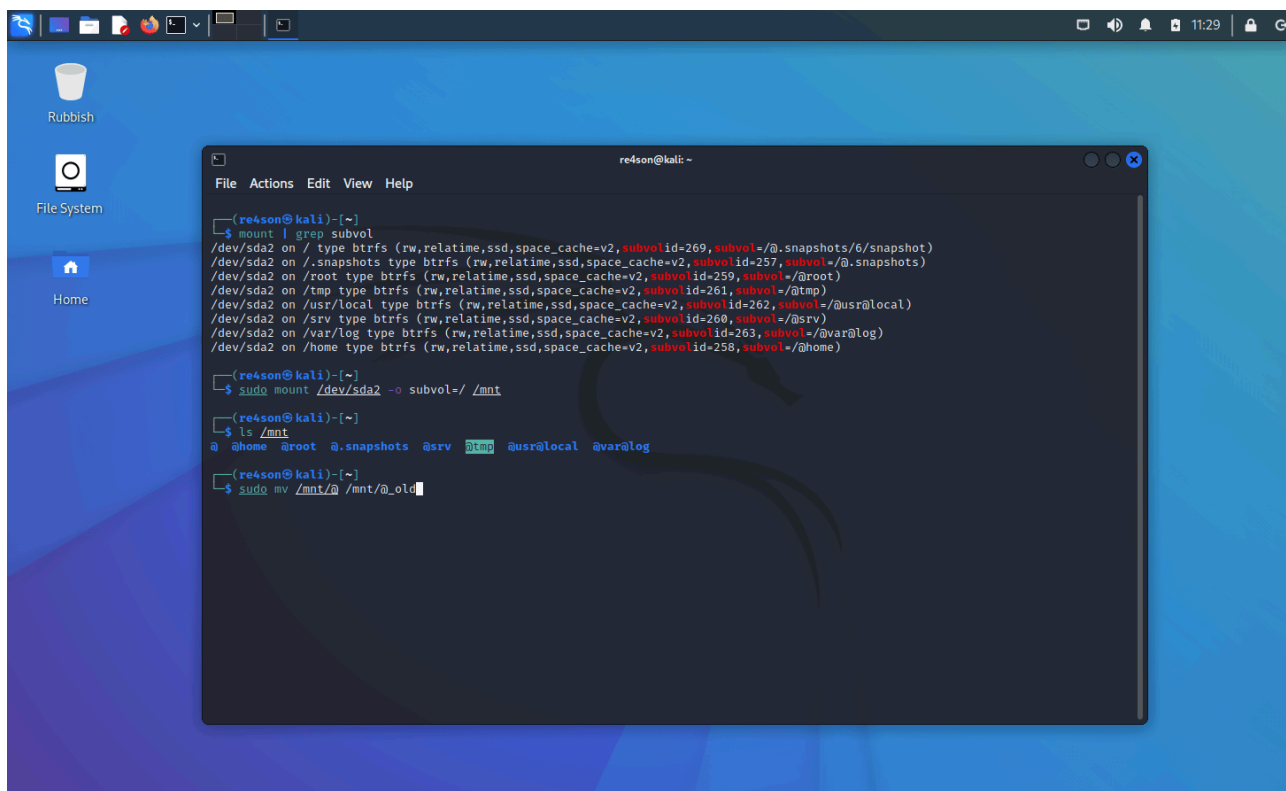
```
sudo mount /dev/sda2 -o subvol=/ /mnt
```

If we list the content of that partition we can see all the subvolumes, including the one containing our snapshots:



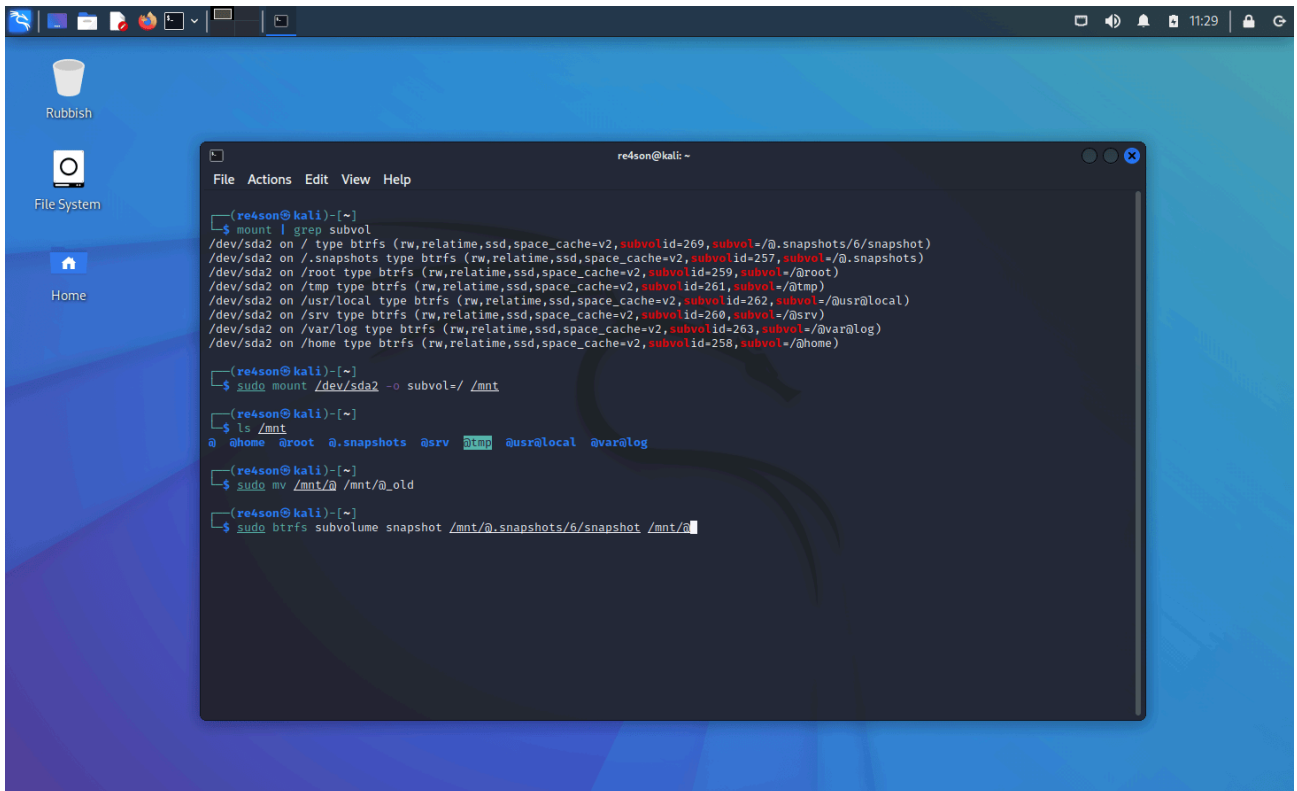
2. Before we replace the current root with our snapshot, let's move "@" away just to be safe:

```
sudo mv /mnt/@ /mnt/@_old
```



1. Now we can create a read-write snapshot of the current read-only snapshot (in this example we booted into the read-only snapshot "6" according to the query above):

```
sudo btrfs subvolume snapshot /mnt/@.snapshots/6/snapshot /mnt/@
```



And that's all there is to it, now we can reboot as if nothing ever happened to our beautiful machine:

References

- [Btrfs Wiki](#)
- [Btrfs Debian site](#)
- [Snapper](#)
- [Snapper-GUI](#)
- [grub-btrfs](#)

Updated on: 2025-Jun-18 Author: [re4son](#)